

Hesse-Activity-1

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Research Interest: Wildfires in Canada's Boreal Forest

My main research interest is tracking the impacts of climate change on post-wildfire forest regrowth in Canada's boreal forest over several decades. The *hypothesis is that increased atmospheric CO₂ will lead to greater rates of evapotranspiration, and therefore, faster regrowth and recovery.* I will be assessing the change in *3 key metrics* over several decades: **rate of canopy height regrowth, rate of biomass regrowth, and rate of albedo recovery.** The study will average forest recovery with fire years dating from the mid-1950s through to the early 2000s. Satellite data will be used to make these assessments where available, while machine learning will be used to simulate forest conditions prior to the launch of Landsat 1. *The goal of this research is to determine if wildfires have had a net effect of warming (due to the release of CO₂ during burning) or cooling (biophysical mechanisms, such as increasing albedo from snow cover and deciduous plant regrowth) on the global climate.* This research is of *political importance* with growing tensions with the USA, as they are currently accusing Canada's wildfires of global pollution, despite the net impact of these wildfires on global climate remaining uncertain.

Favourites

Favourite Music

1. "Cherry Wine" by Hozier
2. "Bobby" by Phoebe Bridgers
3. "From Eden" by Hozier
4. "Rich Girl" by Hall & Oates
5. "Lithonia" by Childish Gambino

Favourite Equation

$$\frac{\partial u}{\partial t} + 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

My favourite equation is the *Korteweg-de Vries (KdV)* equation, which is used to model nonlinear dispersive waves. In 2021, I was awarded an NSERC USRA to work with another student at Brock University to find multi-soliton solutions to the KdV-1 flow equation, under the supervision of Dr. Stephen Anco.

Favourite Artists

Name	Achievements
Hozier	Grammy nominated for Song of the Year in 2015, for song “ <i>Take Me to Church</i> ”
Phoebe Bridgers	Won 4 Grammy awards in 2024 for Best Alternative Music Album , Best Rock Performance , Best Rock Song and Best Pop Duo/Group Performance
Hall & Oates	Inducted into the Rock and Roll Hall of Fame in 2014
Childish Gambino	Won 5 Grammy awards (4 in 2019, 1 in 2018) for Record of the Year , Song of the Year , Best Rap/Sung Performance , Best Music Video , and Best Traditional R&B Performance
Olivia Rodrigo	Won 3 Grammy awards in 2022 for Best New Artist , Best Pop Vocal Album , and Best Pop Solo Performance

A Chunk of Code

```
palette <- c("lightskyblue1", "cornflowerblue", "royalblue3", "blue3", "darkblue")

barplot(
  height = VADeaths,
  beside = T,
  col = palette,
  ylim = c(0,100),
  main = "Death Rates in Virginia, by Demographic, 1940",
  xlab = "Demographic",
  ylab = "Rate of Deaths (of 1000 people)",
  legend.text = rownames(VADeaths),
  args.legend = list(title = "Age Range at Time of Death", x = 20, horiz = T, cex = 0.75)
)
```

